7.

. In the Bohr model of ground-state helium, each electron circles the nucleus, with equal radii and angular momenta , where is the Planck constant.

Ignore electron interaction. Find first ionization energy (eV): energy needed to move one electron to infinity.

The measured ionization energy is . Include electron interaction in this model. Assume the two electrons always occupy opposite ends of a diameter through the nucleus. Find orbit radius , ground energy , ionization energy . Compare the predicted first ionization energy with experiment.

The mass is is light speed;

is Coulomb's constant and is unit charge.

. The photo is a cloud-chamber track, in the paper plane. In its middle a lead plate is normal to paper. A steady uniform field goes into it. The particle charge magnitude is

. The track radius below is , and above is . In lead, the track makes to the normal; the lead is thick; use and

. Ignore gas drag on the particle.

State the direction of motion and sign of charge.

Find the average force in N on the plate while the particle passes through.

Instead, an accelerator pulse of flux hits the plate. One pulse lasts . Find the average force in N on the plate while the pulse passes through, and the heat in J absorbed by the plate during it.